Kittiwut Khongkaeo

Embedded Firmware and LoRaWAN Engineer

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SUMMARY

Embedded engineer with about eight years of firmware work on ESP32 and STM32, mostly C/C++ on FreeRTOS. Most of that time has gone into LoRaWAN systems, from the gateway firmware up to ChirpStack, MQTT and a ThingsBoard dashboard, and into getting them working on site. Looking for remote firmware, RF, UAV/MANET or field deployment work.

EXPERIENCE

Southern IoT Co., Ltd. 2021 – Present

Founder; IoT System Architect & Chief Engineer · Phatthalung, Thailand

- Started the company to design and install LoRaWAN systems in southern Thailand. We are a Seeed Studio partner and distributor.
- Wrote our own gateway firmware so one gateway serves Helium, TTN and ChirpStack at the same time.
- Built a LoRaWAN water metering system in Phatthalung. Residents check usage and pay bills online, and the operator can spot leaks.
- Wrote a LoRa GPS tracker from scratch and field-tested it on three rally cars in the RAAT Thailand Rally Championship.
- Started and led The People's Network, a free public LoRaWAN network started in Nov 2023 after NT announced the end of its LoRaWAN service. It grew to 100+ gateways and expanded to Malaysia and the UK.
- Running two PoCs now: CO2 monitoring for carbon credits and racing telemetry.
- Gave a talk at a depa (Digital Economy Promotion Agency) Smart City Thailand webinar in Jan 2024 on community sensor networks.

ConnexInsights Co., Ltd. 2025 – 2026

IoT Architecture and Design

- Led LoRaWAN network work for government utilities and state enterprises.

Defense technology company, Thailand (name confidential) 2024 – Present

Defense Systems Developer

- Worked on weapon systems design, from early design through development.
- Developed a command and control (C2) system for short-range air defense (SHORAD).
- Developing strike and reconnaissance UAVs with autonomous functions, covering the embedded software and firmware.
- Wrote firmware that turns Wi-Fi HaLow devices into MANET mesh radios for military and disaster comms, used with ATAK.
- Built a long-range UAV control and telemetry link based on ExpressLRS, flight-tested and listed at TRL 6.
- Connected LoRaWAN trackers and sensors to TAK/ATAK.

Earlier work: aviation and oilfield maintenance 2013 – 2020

Aircraft line maintenance at Lion Air, AirAsia and Asia Atlantic Airlines (Boeing 767), flight simulator maintenance at WM Simulator, and M/LWD drilling tool maintenance in the oilfield. Holds company maintenance authorizations from ANA and Asia Atlantic Airlines.

SELECTED PROJECTS

- **Mesh LoRaWAN gateways.** Multi-hop LoRaWAN gateway mesh in Bangkok, used for metering, asset tracking and weather stations, and to cut backhaul costs.
- **CO2 monitoring for carbon credits.** Rewrote CO2 sensor firmware and recalibrated the sensors. Part of this work is with Krumap and Sasin School of Management, Chulalongkorn University (2025 – present).
- **Rust packet forwarder.** Ported the gateway driver and packet forwarder from C to Rust and tested it in the field.
- **Multi-GNSS board.** STM32 board that combines 8 GNSS receivers into one position for the flight computer.

- **Flood response.** During the late-2025 southern floods, used Meshtastic and ATAK to track boats and rescue teams in Krabi, Trang and Phatthalung.

OPEN SOURCE

- **Meshtastic firmware.** Merged contribution (Apr 2026): Thailand radio settings to match NBTC rules.
- **ExpressLRS.** Merged contribution (Aug 2026): added Thailand's frequency band.
- **OpenTAKServer.** Merged contribution (Mar 2026): full Thai translation.
- **sx1302_hal v3.0.0.** Updated Semtech's gateway driver for current Raspberry Pi OS.
- **SenseCAPM1-Chirpstack.** Scripts and guide for running a SenseCAP M1 as a private ChirpStack gateway.

SKILLS

Firmware: C, C++, Rust, Python; ESP32, STM32, Raspberry Pi; FreeRTOS; embedded Linux (Debian, Pi OS)

Radio / Networks: LoRa and LoRaWAN, Semtech SX1276 / SX1302 / SX1303, Meshtastic, ExpressLRS, Wi-Fi HaLow MANET, Reticulum, mioty

IoT platforms: ChirpStack, Helium, TTN / The Things Stack, MQTT (Mosquitto, EMQX), Node-RED, ThingsBoard, RS485 field devices

Tools: Linux, Docker, Kubernetes, Git / GitHub, TAK / ATAK / WinTAK, OpenTAKServer, TensorFlow, PyTorch

EDUCATION

Civil Aviation Training Center (Thailand) Certified 17 Sep 2011
ICAO-approved course, Aircraft Instruments (Aircraft Maintenance Technician, Type I)
Type ratings: B737NG, A330, B767

LANGUAGES

Thai, English